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Nomenclature

Greek Symbols

π $\simeq 3.14\dots$

Acronyms / Abbreviations

AMO Atlantic Multidecadal Oscillation

BSISO Boreal Summer Intraseasonal Oscillation

ENSO El Niño–Southern Oscillation

MJO Madden-Julian Oscillation

NAO North Atlantic Oscillation

PC Principal Component

PDO Pacific Decadal Oscillation

VC Vector of Coefficients

Theoretical Foundations

1.1 | Principal Component Analysis

Principal Component Analysis (PCA) is a statistical technique used to reduce the dimensionality of a dataset while preserving most of the variance. This is achieved by identifying the principal axes, which are the directions in which the data varies the most. These directions correspond to the eigenvectors of the covariance matrix of the data, with the associated eigenvalues representing the magnitude of variance along these directions. PCA is particularly useful for dimensionality reduction, data compression, feature extraction and visualization of high-dimensional datasets.

The origins of PCA can be traced back to Pearson [62] in 1901 and Hotelling [41] in 1936. It gained significant traction in meteorology through the work of Lorenz [55] in 1956, who introduced the technique under the term *empirical orthogonanal function* (EOF) analysis¹. The method has since been extensively applied in meteorology and climate science to identify dominant modes of variability in the atmosphere and oceans, as discussed in numerous textbooks [e.g. 23, 35, 44, 63, 71, 80].

In this section, we derive PCA from the maximum variance formulation², as outlined by Bishop [4, pp. 561-563]. This approach involves projecting the data onto a lower-dimensional linear space, known as the principal subspace, to maximize the variance of the projected data. We will then explore the limitations of PCA, particularly in the context of climate science, and discuss common strategies to address these limitations.

nicrie: make link to Karhunen-Loève expansion

¹Although EOF analysis remains the preferred term in meteorology and climate science, we will use the more commonly recognized term, PCA.

²Hotelling [41] originally proposed this formulation. An alternative derivation, known as the minimum error formulation by Pearson [62], seeks a linear projection that minimizes the average projection error, defined as the mean squared distance between the original data points and their projections.

1.1.1 Derivation from Maximum Variance Formulation

Consider a data set represented as a complex-valued, two-dimensional $N \times D$ matrix $\mathbf{Z}$, where N denotes the number of observations (or samples) and D the number of variables (or features). We will refer to the n^{th} observation as the D -dimensional vector $\mathbf{z}_n$. For simplicity, we assume the data is centered, meaning the mean of each variable is zero, i.e. $\bar{\mathbf{z}} = \frac{1}{N} \sum_{n=1}^N \mathbf{z}_n = \mathbf{0}$. The goal is to project the data onto an M -dimensional space ($M < D$) while maximizing the variance of the projected data.

To start, consider a one-dimensional projection ($M = 1$). We define the direction of this space using a D -dimensional unit³ vector $\mathbf{v}_1$, such that $\mathbf{v}_1^H \mathbf{v}_1 = 1$, where $\mathbf{v}_1^H$ denotes the conjugate transpose of $\mathbf{v}_1$. The data points $\mathbf{z}_n$ are then projected onto this direction, $\mathbf{v}_1^H \mathbf{z}_n$, and the variance of the projected data is

$$\frac{1}{N-1} \sum_{n=1}^{N-1} (\mathbf{v}_1^H \mathbf{z}_n - \mathbf{v}_1^H \bar{\mathbf{z}}) (\mathbf{v}_1^H \mathbf{z}_n - \mathbf{v}_1^H \bar{\mathbf{z}})^H = \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1, \quad (1.1)$$

where $\mathbf{S}_{ZZ}$ is the covariance matrix of $\mathbf{Z}$ given by

$$\mathbf{S}_{ZZ} = \frac{1}{N-1} \sum_{n=1}^N \mathbf{z}_n \mathbf{z}_n^H = \frac{1}{N-1} \mathbf{Z}^H \mathbf{Z}. \quad (1.2)$$

The goal is to find the vector $\mathbf{v}_1$ that maximizes this variance, formulated as:

$$\underset{\mathbf{v}_1}{\operatorname{argmax}} \quad \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 \quad \text{subject to} \quad \mathbf{v}_1^H \mathbf{v}_1 = 1. \quad (1.3)$$

To solve this optimization problem, we maximize the projected variance $\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1$ with respect to $\mathbf{v}_1$. To enforce the normalization constraint $\mathbf{v}_1^H \mathbf{v}_1 = 1$, a Lagrange multiplier λ_1 is introduced, leading to the unconstrained maximization of:

$$\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 + \lambda_1 (1 - \mathbf{v}_1^H \mathbf{v}_1). \quad (1.4)$$

By setting the derivative with respect to $\mathbf{v}_1$ to zero, we find a stationary point when:

³This normalization is for convenience and not necessary, since we are only interested in the direction of $\mathbf{v}_1$, not its magnitude.

$$\mathbf{S}_{ZZ}\mathbf{v}_1 = \lambda_1\mathbf{v}_1. \quad (1.5) \quad 1$$

This indicates that $\mathbf{v}_1$ must be a eigenvector of $\mathbf{S}_{ZZ}$, with λ_1 being the corresponding real-valued⁴ eigenvalue. The variance $\mathbf{v}_1^T \mathbf{S}_{ZZ} \mathbf{v}_1$ reaches its maximum when $\mathbf{v}_1$ aligns with the eigenvector associated with the largest eigenvalue⁵ λ_1 . The eigenvector is referred to as the first **vector of coefficients**⁶ (VC) of the data set. 2 3 4 5

To find additional VCs, we iteratively select new directions that maximize the projected variance while being orthogonal to the previously chosen directions. For an M-dimensional projection space, the optimal linear projection that maximizes the variance of the projected data is defined by the M eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_M$ of the covariance matrix $\mathbf{S}_{ZZ}$, corresponding to the M largest eigenvalues $\lambda_1, \dots, \lambda_M$. This result can be derived using proof by induction, leading to the concise expression: 6 7 8 9 10 11

$$\mathbf{S}_{ZZ}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}, \quad (1.6) \quad 12$$

where $\mathbf{V}$ is unitary, i.e. $\mathbf{V}^T \mathbf{V} = \mathbf{I}_M$. 13

Thus, the eigenvectors of $\mathbf{S}_{ZZ}$ emerge as the new coordiante system "along" regularities in the data. For cases where these regularities are not aligned in unitary directions, see Section 1.1.2. 14 15

According to the spectral theorem for normal matrices [e.g. 40, p. 133], the hermitian covariance matrix $\mathbf{S}_{ZZ}$ can be decomposed into the sum of the outer products of the eigenvectors $\mathbf{v}_i$ and the corresponding real-valued eigenvalues λ_i : 16 17 18

$$\mathbf{S}_{ZZ} = \sum_{i=1}^M \lambda_i \mathbf{v}_i \mathbf{v}_i^H. \quad (1.7) \quad 19$$

Each eigenvalue λ_i represents the amount of variance captured by the corresponding component, typically expressed as a fraction γ_i of the total variance: 20 21

⁴This follows from the property of the covariance matrix being hermitian.

⁵This can easily verified by left-multiplying Equation (1.5) with $\mathbf{v}_1^H$ and making use of the normalization condition.

⁶Naming conventions can vary significantly across different scientific domains. For example, in machine learning, the terms principal components and principal component scores refer to the eigenvectors (here $\mathbf{V}$) and their corresponding data projections (here $\mathbf{P}$) [e.g. 4]. In statistics, these are often called vectors of coefficients/loadings and principal components [e.g. 44], while in climate science and meteorology, they are known as empirical orthogonal functions and principal components/expansion coefficients [e.g. 71]. In this thesis, I aim to maintain consistency with the terminology commonly used in statistics.

$$\gamma_i = \frac{\lambda_i}{\sum_{i=1}^M \lambda_i} \quad (1.8)$$

Projecting the data $\mathbf{Z}$ onto the M -dimensional subspace thus provides a lower dimensional representation, called the principal components (PCs),

$$\mathbf{P} = \mathbf{Z}\mathbf{V}, \quad (1.9)$$

which also enables the reconstruction of the original data from the projected data:

$$\mathbf{Z} = \mathbf{P}\mathbf{V}^H. \quad (1.10)$$

However, calculating the eigenvector decomposition is not always computationally efficient. A more efficient method involves using Singular Value Domposition (SVD) [e.g. 40, p. 150], which decomposes the data matrix as

$$\mathbf{Z} = \mathbf{U}'\mathbf{\Sigma}\mathbf{V}'^H, \quad (1.11)$$

where $\mathbf{U}'$ and $\mathbf{V}'$ are unitary matrices, $\mathbf{U}'^H\mathbf{U}' = \mathbf{V}'^H\mathbf{V}' = \mathbf{I}_M$, and $\mathbf{\Sigma}$ is a diagonal matrix containing the singular values σ_i of $\mathbf{Z}$. The covariance matrix $\mathbf{S}_{ZZ}$ can then be expressed in terms of the SVD as

$$\mathbf{S} = \frac{1}{N-1}\mathbf{Z}^H\mathbf{Z} = \frac{1}{N-1}\mathbf{V}'\mathbf{\Sigma}^H\mathbf{U}'^H\mathbf{U}'\mathbf{\Sigma}\mathbf{V}'^H = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^H, \quad (1.12)$$

where, by comparison with (1.6), $\mathbf{V} = \mathbf{V}'$ and $\mathbf{\Lambda} = \frac{1}{N-1}\mathbf{\Sigma}^T\mathbf{\Sigma}$. This establishes a direct relationship between the eigenvalues of the covariance matrix and the singular values of the data matrix:

$$\lambda_i = \frac{\sigma_i^2}{N-1}. \quad (1.13)$$

Comparing (1.10) with (1.11) shows that the PC scores $\mathbf{P}$ can be normalized

$$\mathbf{U} \equiv \mathbf{U}' = \mathbf{P}\mathbf{\Sigma}^{-1} = \frac{1}{(N-1)^{1/2}} \mathbf{P}\mathbf{\Lambda}^{-1/2}, \quad 1$$

or alternatively $\mathbf{P} = (N-1)^{1/2} \mathbf{U}\mathbf{\Lambda}^{1/2}. \quad (1.14) \quad 2$

Thus, $\mathbf{U}^H \mathbf{U} = \mathbf{I}_M$ ensures that the PC scores $\mathbf{U}$ are unitary and uncorrelated. 3

A common scaling method for the vector of coefficients $\mathbf{V}$ involves scaling by the square root of the corresponding eigenvalues, referred to as **loadings**: 4
5

$$\tilde{\mathbf{V}} = \mathbf{V}\mathbf{\Lambda}^{1/2}. \quad (1.15) \quad 6$$

Although not strictly unitary, these loadings form a generalized unitary matrix, as shown by 7

$$\tilde{\mathbf{V}}^H \tilde{\mathbf{V}} = \mathbf{\Lambda}. \quad (1.16) \quad 8$$

Using this scaling, the Equation 1.10 can be expressed as 9

$$\mathbf{X} = (N-1)^{1/2} \mathbf{U}\tilde{\mathbf{V}}^T. \quad (1.17) \quad 10$$

Computational Considerations The computational cost of performing eigendecomposition on the covariance matrix scales with $O(D^3)$, making it impractical for modern datasets, such as those in climate science, where typical data sizes are on the order of $D = O(10^6)$ or even larger. For example, ERA5 [13], a widely used reference for state-of-the-art reanalysis climate data, distributes a single climate variable on a regular 0.25° grid, resulting in $D = 1\,036\,800$ grid cells. 11
12
13
14
15

In practice, more efficient algorithms, such as SVD or power methods [31], have been employed traditionally employed. Recently, randomized low-rank approximation techniques [24, 34] have further reduced computational complexity to $O(ND \log(M) + (N+D)M^2)$. These advancements offer feasible alternatives for handling large, out-of-memory datasets (see Table 1.1). 16
17
18
19

Shortcomings Despite its widespread use and versatility, PCA has several limitations and potential drawbacks that should be considered: 20
21

- 1 **1. Linearity Assumption:** PCA assumes a linear structure in the data, which may not
2 hold in complex systems such as climate data, where nonlinear relationships can be
3 significant [48, 70].
- 4 **2. Interpretability of VCs:** The orthogonality of the vectors of coefficients is a mathematical
5 construct that may not always correspond to the physical reality of the data, potentially
6 leading to misinterpretations [67].
- 7 **3. Variance Focus:** Focusing on directions of maximum variance may not always capture
8 the most relevant features for a specific application, potentially critical but less variable
9 information [27].
- 10 **4. Static Nature:** As a static method, PCA does not account for temporal dynamics or
11 changes over time, which is particularly limiting in climate science, where time evolution
12 is a key aspect.
- 13 **5. Sensitivity to Outliers:** PCA can be sensitive to outliers, as extreme values can dispro-
14 portionately influence the VCs, leading to misleading results [16].
- 15 **6. Sensitivity to Scaling:** The results are sensitive to the scaling of the input data, meaning
16 that variables with larger ranges can dominate the VCs unless the data is appropriately
17 standardized.
- 18 **7. Loss of Information:** While dimensionality reduction is often the goal, PCA inherently
19 discards some information, which could lead to the loss of important details crucial for
20 accurate analysis and interpretation in specific applications.

21 Over the past few decades, numerous modifications and extensions have been proposed to
22 address these limitations, especially in the context of climate science. For instance, to address
23 the linearity assumption, various forms of kernel PCA [10, 70, 78] have been developed. For
24 improved interpretability, rotated [67] and sparse [45] PCA methods have been proposed. Ro-
25 bust PCA [16] offers a more resilient approach to handling sensitivity to outlier. Furthermore,
26 extended [8, 77] and complex [1, 38] PCA methods have been introduced to account for tempo-
27 ral dynamics. In addition, eigenmethods like Optimal Persistence Analysis (OPA) [21, 22] and
28 Principal Oscillation Pattern (POP) analysis [36] provide alternative approaches independent
29 to variance maximization to understanding dominant patterns in physical climate fields.

30 In the following sections, we will explore some of these shortcomings and proposed solutions
31 in greater detail. Specifically, we will discuss how additional rotation of the PCA solution can
32 enhance physical interpretability (Section 1.1.2) and how complexifying the data can provide
33 insights into time evolution patterns (Section 1.1.3).

Table 1.1 Computational costs of different algorithms used to compute either full or approximative M -rank solutions for PCA for a real, dense, in-memory matrix with dimensions $N \times D$. Compiled from Bishop [4] and Halko *et al.* [34].

Algorithm	Complexity
Eigenvector decomposition	$O(D^3)$
Full SVD	$O(ND \min(N, D))$
Power method	$O(MD^2)$
Truncated SVD	$O(NDM)$
Randomized SVD	$O(ND \log(M) + (N + D)M^2)$

1.1.2 Sparse PCA via Rotation

PCA is widely used for dimensionality reduction, as it maximizes the explained variance and provides orthogonality, making it optimal in a least-squares sense [4]. It effectively identifies principal axes that capture the covariance among variables, allowing for the examination of interrelationships between stations, grid points or variables. However, PCA is not ideal for identifying the cause of variation or deriving physical modes of variability [44, 51, 67].

One limitation is that real-world processes do not necessarily adhere to the orthogonality imposed by PCA's eigenvectors. Additionally, the dense nature of these VCs – where all elements are non-zero – can make interpretation challenging. In the context of climate and environmental science, where variables are often represented across spatial grids, these limitations can lead to recognizable but artificial patterns known as Buell patterns [9, 50, 67]. These patterns typically start with a positive component and evolve into increasingly complex structures, such as bipole or tripole patterns (Figure 1.1). Such issues highlight the disconnect between standard PCA and underlying physical processes [67], prompting the development of alternative methods that produce sparse VCs, characterized by many zero or near-zero elements. This sparsity – which necessarily implies the loss of orthogonality for at least one of the VCs and PC scores – can enhance interpretability by localizing the patterns and reducing the density of the VCs.

In climate science, the most common approach to achieving sparse PCA is through rotation techniques [39, 66, 67], a concept borrowed from factor analysis [cf. 44, pp. 269]. In this "factor analysis viewpoint" of PCA [44, pp. 150-161], a linear transformation is applied to a subset of the VCs to derive a new set of rotated VCs that are "simpler" in a heuristically defined manner, ideally aligning better with the underlying physical processes.

As noted in Equation (1.17), once the orthogonality constraint is relaxed, PCA solutions exhibit inherent indeterminacy, allowing any non-singular transformation $\mathbf{R}$ to satisfy the PCA formulation [e.g. 67, p. 305]:

$$\mathbf{Z} = (\mathbf{N} - 1)^{1/2} \mathbf{U} \tilde{\mathbf{V}}^H = (\mathbf{N} - 1)^{1/2} \mathbf{U} \mathbf{R} \mathbf{R}^{-1} \tilde{\mathbf{V}}^H. \quad (1.18)$$

Rotated PCA exploits this indeterminacy by choosing $\mathbf{R}$ to optimize a simplicity criterion f , resulting in rotated loadings⁷ $\tilde{\mathbf{V}}_R = \tilde{\mathbf{V}} \mathbf{R}$ that maximize f :

$$\underset{\mathbf{R}}{\operatorname{argmax}} f(\tilde{\mathbf{V}}_R). \quad (1.19)$$

There are two main categories of rotation: *orthogonal*⁸ and *oblique*. In orthogonal rotation, $\mathbf{R}$ is a unitary matrix, preserving the uncorrelated nature of the PCs since $\mathbf{U}_R^H \mathbf{U}_R = \mathbf{R}^H \mathbf{U}^H \mathbf{U} \mathbf{R} = \mathbf{I}_M$. In oblique rotation, $\mathbf{R}$ is a general non-singular matrix, resulting in correlated PCs, as $\mathbf{U}_R^H \mathbf{U}_R = \mathbf{R}^H \mathbf{R}$. Regardless of the method, the rotated loadings are not orthogonal,

$$\tilde{\mathbf{V}}_R^H \tilde{\mathbf{V}}_R = \mathbf{R}^H \tilde{\mathbf{V}}^H \tilde{\mathbf{V}} \mathbf{R} \stackrel{(1.16)}{=} \mathbf{R}^H \mathbf{A} \mathbf{R} \neq \mathbf{I}_M, \quad (1.20)$$

which is crucial for achieving a more interpretable solution. Various simplicity criteria f have been proposed [see e.g. 67], with most focusing on making the rotated loadings more localized and physically interpretable. In practice, the specific choice of simplicity criterion often has little impact on the results [67].

Varimax Orthogonal Rotation The most widely used orthogonal rotation technique is the varimax criterion [46], which aims to simplify the VCs by maximizing the variance of the squared loadings,

$$f(\tilde{\mathbf{V}}_R) = \sum_{m=1}^M \left(p \sum_{j=1}^P b_{jm}^2 - \left(\sum_{j=1}^P b_{jm} \right)^2 \right) \quad (1.21)$$

⁷In principle, it is possible to rotate the unitary matrix of coefficients $\mathbf{V}$ instead of the loadings matrix $\tilde{\mathbf{V}}$ [44, 58]. However, this approach is seldom used in practice. The primary reason is that rotating the unitary coefficient matrix preserves the orthogonality of the VCs but results in the loss of orthogonality among the PCs, which does not address issues like the Buell patterns. Moreover, this method does not consider the relative importance of the spatial patterns, as indicated by the eigenvalues, making the results of the rotation process highly sensitive to the selection of the number of components to rotate.

⁸In the broader context of complex data, this would be referred to as a unitary rotation. However, since factor analysis has primarily been applied to real-valued data, the terms orthogonal and oblique rotations have become the standard. To avoid confusion, I use these historical terms while acknowledging that they also encompass the more general case of complex data.

1.1 Principal Component Analysis

9

where b_{jm} are the squared norm of the rotated loadings $|\tilde{v}_{R,jm}|^2/h_j^2$ normalized by the communalities $h_j^2 = \sum_{m=1}^M |\tilde{v}_{R,jm}|^2$ of variable j . The varimax rotation pushes the elements of $\tilde{\mathbf{V}}_R$ towards either 0 or 1, creating a sparse solution with primarily large or small values, which is particularly useful for identifying localized physical processes, such as atmospheric or oceanic teleconnections.

Promax Oblique Rotation The promax criterion [37], a generalization of the varimax criterion, is one of the most common oblique rotation techniques. It is based on the premise that an orthogonal simple structure solution, such as varimax, is often close to an optimal oblique solution. The promax method constructs a hypothetical oblique target matrix $\mathbf{T}$ by performing an oblique Procrustes transformation [42] on the varimax-rotated loadings matrix,

$$\mathbf{T} = \tilde{\mathbf{V}}_R \mathbf{R} + \mathbf{E}, \quad (1.22)$$

where $\mathbf{R}$ is the transformation matrix and $\mathbf{E}$ is a matrix of residuals. The normal equation for solving the complex Procrustes problem is given by:

$$\mathbf{R} = (\tilde{\mathbf{V}}_R^H \tilde{\mathbf{V}}_R)^{-1} \mathbf{T}^H \tilde{\mathbf{V}}_R. \quad (1.23)$$

To define the target matrix $\mathbf{T}$, the promax criterion increases the discrimination between high and low loadings by raising the loadings to a power q while retaining the original sign,

$$t_{jm} = \frac{|a_{jm}|^{q+1}}{a_{jm}}, \quad (1.24)$$

where q controls the degree of obliqueness and a_{jm} are the elements of the max-normalized, varimax-rotated loadings matrix $\mathbf{A}$, defined as

$$\mathbf{A} = \tilde{\mathbf{A}} \mathbf{K}^{-1}, \quad (1.25)$$

$$\tilde{\mathbf{A}} = \mathbf{H}^{-1} \tilde{\mathbf{V}}_R, \quad (1.26)$$

with $\mathbf{K}$ being a diagonal matrix containing the maximum absolute values of each column of $\tilde{\mathbf{A}}$, $a_m = \max_m |A_{pm}|$, and $\mathbf{H}$ a diagonal matrix with the square root of the communalities h_j for the j^{th} variable.

Application and Criticism Both orthogonal and oblique rotation techniques, particularly varimax and promax, have been extensively used since the 1980s to identify stationary modes of variability in climate data [e.g. 2, 30, 39, 43, 49, 54, 59]. Notable climate oscillations, such as the North Atlantic Oscillation (NAO), have been defined based on varimax-rotated patterns of loadings [2]. These techniques also help identify "ghost patterns", artificial patterns identified by PCA that are not physically meaningful [43, 53].

Rotated modes offer several advantages, such as greater robustness against temporal and spatial subsampling [18] compared to unrotated modes. They also effectively mitigate the statistical artifacts associated with Buell patterns, resulting in more realistic representations of the underlying autocorrelation structure and thus increased interpretability [53, 67, 68]. Moreover, rotation techniques have been successfully integrated into other PCA extensions dealing with the non-linear and non-stationary nature of climate oscillations [10, 33, 38].

However, rotation techniques are not without criticism, mainly due to the ad-hoc nature of the various simplicity criteria [cf. 44, pp. 270-274]. This has led to the exploration of alternative approaches to enforce sparsity. Notably, Jolliffe *et al.* [45] introduced the Simplified Component Technique - LASSO (SCoTLASS), a sparse PCA variant based on the least absolute shrinkage and selection operator (LASSO), originally from regularized regression [73]. This method imposes a constraint on the ℓ_1 norm of the vector of loadings, resulting in some loading coefficients being exactly zero.

Despite its advantages, SCoTLASS has drawbacks, particularly when the number of variables D is much larger than the number of observations N . In such cases, SCoTLASS can only select at most N non-zero elements in the loading vectors, which is not ideal in climate science where often $D \gg N$. Additionally, since SCoTLASS solutions cannot be obtained via SVD and require a numerical approach using projected gradient methods [74], the computational demand is typically higher than for (rotated) PCA. While this method has inspired many other sparse PCA techniques based on penalties or constraints (usually ℓ_0 or ℓ_1) [see e.g. 26, 81, 82], these formulations are often complex and require algorithmic innovations to avoid issues like local optima and ensure convergence. They may also face challenges with memory limitations, slow computation speeds and limited effectiveness in modelling the data [15, 32].

It is likely for these reasons, that rotated PCA remains one of the most commonly used techniques for achieving sparse PCA solutions in climate science. It is computationally efficient, straightforward to implement and provides a more interpretable understanding of the data compared to standard PCA.

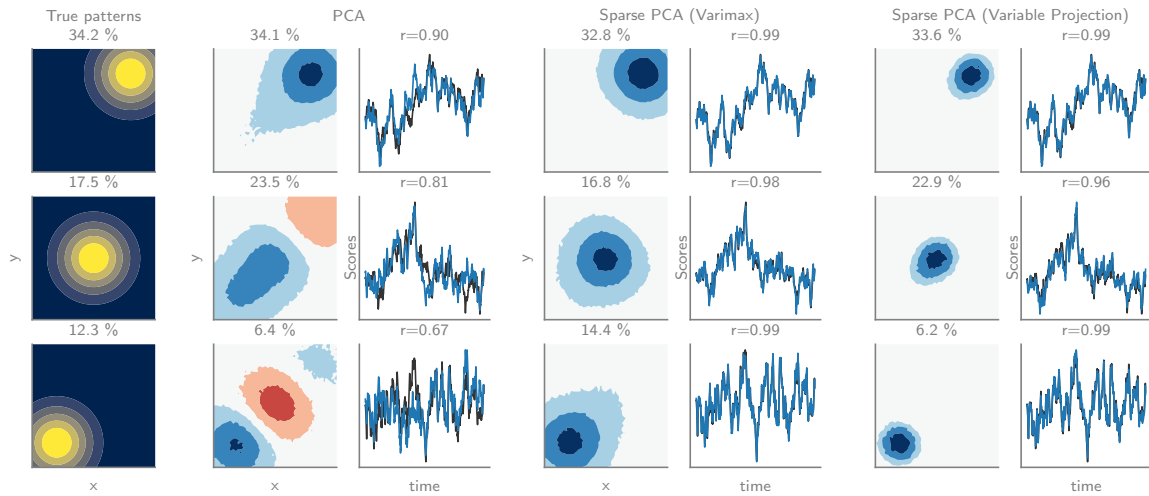


Fig. 1.1 Comparison of the first three modes obtained from PCA vs. Varimax-rotated PCA for a synthetic data set. (Left) Original data set with three different, uncorrelated autoregressive processes. (Middle) PCA solution. (Right) Varimax-rotated PCA solution.

1.1.3 Hilbert PCA

While PCA efficiently identifies the principal axes along which the covariance among all variables is distributed, and rotated PCA provides a more sparse solution related to underlying physical processes, both methods are inherently static. They do not account for temporal dynamics or changes over time, as they only consider the current state of the physical field. This limitation is significant in climate science, where time evolution is crucial. For instance, standard PCA does not differentiate between the order of data points over time, resulting in the same eigenvectors and eigenvalues regardless of temporal permutations.

This static nature poses a challenge for capturing oscillatory patterns, such as QBO, MJO, ENSO and BSISO, which involve spatial patterns propagating over time. These phenomena cannot be adequately captured by standard PCA.

Hilbert Transform as a Momentum Operator

To address this limitation, one approach is to model both the current state of the climate and its tendency or 'momentum', analogous to the Hamiltonian principle in mechanics, where a system's future is described by first-order differential equations for location (state) and momentum (tendency) [cf. 71, p. 353]. In classical mechanics, propagating features are represented using canonical coordinates: each state variable is augmented by a momentum variable, allowing oscillations to be depicted as rotations in a two-dimensional phase space.

1 From a data perspective, this augmentation can be achieved via the Hilbert transform ($\mathcal{H}$). For
 2 a real time series $x(t)$, the Hilbert transform $x'(t)$ is defined as

$$3 \quad x'(t) \equiv \mathcal{H}\{x\}(t) = \frac{1}{\pi} \text{p.v.} \int_{-\infty}^{\infty} \frac{x(\tau)}{\tau - t} d\tau, \quad (1.27)$$

4 where p.v. denotes the Cauchy principal value ensuring convergence despite the singularity at
 5 $\tau = t$. The Hilbert transform emphasises the local properties of $x(t)$ by convolution with $\frac{1}{\pi t}$. In
 6 the frequency domain, it is related to the Fourier transform $\mathcal{F}$ by

$$7 \quad \mathcal{F}\{y\}(\omega) = -i \operatorname{sgn}(\omega) \mathcal{F}\{x\}(\omega) \quad (1.28)$$

8 where the imaginary unit i indicates a phase shift of $\pi/2$ for the frequency components ω of x ,
 9 placing x' in quadrature to x . This provides an intuitive interpretation of the Hilbert transform,
 10 as it acts as a linear filter that alters the phase of the signal's components by 90 degrees without
 11 changing their amplitude.

12 By using the Hilbert transform, a real-valued time series $x(t)$ can be complexified into $z(t) =$
 13 $x(t) + i\mathcal{H}\{x\}(t)$, known as the *analytic signal*, which allows for the computation of the instan-
 14 taneous amplitude and phase of $x(t)$:

$$15 \quad A(t) = |z(t)| = \sqrt{x(t)^2 + \mathcal{H}\{x\}(t)^2}, \quad \phi(t) = \arg(z(t)) = \arctan\left(\frac{\mathcal{H}\{x\}(t)}{x(t)}\right). \quad (1.29)$$

16 Similar results can be obtained for the discrete case [61, pp. 942-956], and this concept can
 17 be extended to multivariate data [e.g. 35, pp. 105-106], defining the analytic signal of a data
 18 matrix $\mathbf{X}$ as

$$19 \quad \mathbf{Z} = \mathbf{X} + i\mathcal{H}\{\mathbf{X}\}, \quad (1.30)$$

20 where $\mathcal{H}\{\mathbf{X}\}$ is the feature-wise Hilbert transform of $\mathbf{X}$.

21 Rasmusson *et al.* [64] introduced the concept of Hilbert PCA using this data augmentation
 22 technique⁹. This method, refined by Barnett [1] and Horel [38], involves applying PCA to
 23 the augmented data matrix $\mathbf{Z}$. By incorporating phase information, Hilbert PCA can capture

⁹Although Rasmusson *et al.* [64] originally termed it *Hilbert Singular Decomposition*, the terms *Hilbert EOF/PCA* and *Complex EOF* analysis are more commonly used in the literature [12, 35, 44, 71].

the direction and speed of wave-like patterns, unlike traditional PCA, which only identifies standing oscillations. The resulting eigenvectors and principal component scores are complex, with the real and imaginary components linked via the Hilbert transform. The variance capture by each component is represented by the real eigenvalues of the complex covariance matrix of Z (see Section 1.1).

The main advantage of Hilbert PCA is the ability to identify and reconstruct propagating waves, which is especially valuable in studying climate phenomena characterized by such patterns, including MJO, ENSO, QBO and BSISO. For instance, Cai & Baines [14] utilized Hilbert PCA to analyze SST in the southern hemisphere, linking the findings to ENSO, while Bürger [12] employed it to develop an index for BSISO. However, to reliably identify oscillatory patterns, sufficiently long time series are required to resolve at least one full oscillation [57]. Additionally, interpreting of the Hilbert transform as representing the instantaneous frequency is valid holds for narrow-band signals; the phase of the Hilbert transform is not well-defined for broadband signals [6, 29]. Moreover, real-time monitoring is challenging with Hilbert PCA because new real-valued data cannot be directly projected onto the existing patterns without knowing or estimating the exact momentum (phase) of the new data, which requires future information (Equation (1.27)). To mitigate this, an approximative real-time (causal) Hilbert transform can be employed [11, 12], offering a practical remedy to the issue of real-time applicability. Finally, Hilbert PCA shares PCA's limitations regarding physical interpretability due to the orthogonality of the VCs. To address this, rotated Hilbert PCA has been proposed to enhance interpretability [5, 39].

1.2 | Maximum Covariance Analysis

In contrast to finding latent variables that maximize covariance within a single dataset, Maximum Covariance Analysis (MCA) aims to identify the latent variables that are common in two sets of data and maximize covariance. For example, in the case of the MJO [56], which is the dominant mode of intraseasonal variability in the tropical atmosphere, characterized by an eastward-moving cell of overturning wind circulation and associated regions of enhanced and suppressed precipitation, MCA helps in findings shared dynamics between different variables. This method allows us to focus on shared variability while ignoring dominant variations associated with only one variable, such as diurnal cycles linked to surface processes absent at higher atmosphere levels.

MCA identifies patterns of maximum covariance between datasets by finding pairs of linear subspaces with high cross-covariance [e.g. 71, p. 321]. Unlike PCA, which operates on a single set of variables, MCA considers two¹⁰ sets of variables and seeks components that exhibit

¹⁰In theory, this can be extended to more than two sets of data [47].

high covariance across datasets. This approach can be formulated as an optimization problem, aiming to find pairs of (complex-valued) coupled patterns $\mathbf{v}_{s,1}$ for datasets $\mathbf{Z}_s$, $s = x, y$, such that the projection maximizes the covariance between the datasets. The solution involves a generalized eigenvector problem, often solved using the Singular Value Decomposition (SVD) of the cross-covariance matrix.

$$\operatorname{argmax}_{\mathbf{v}_{x,1}, \mathbf{v}_{y,1}} \mathbf{v}_{x,1}^H \mathbf{C}_{xy} \mathbf{v}_{y,1} \quad \text{subject to} \quad \|\mathbf{v}_{x,1}\|_2 = \|\mathbf{v}_{y,1}\|_2 = 1, \quad (1.31)$$

where the cross-covariance matrix between the two datasets is given by

$$\mathbf{C}_{xy} = \frac{1}{N-1} \mathbf{Z}_x^H \mathbf{Z}_y. \quad (1.32)$$

The solution to this problem gives rise to a generalized eigenvalue equations similar to that of PCA, which can also be solved by using the SVD of the cross-covariance matrix

$$\mathbf{C}_{xy} = \mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^H \quad (1.33)$$

where the i^{th} column of the left and right singular vectors, $\mathbf{V}_x$ and $\mathbf{V}_y$, contains the pattern vectors $\mathbf{v}_{x,i}$ and $\mathbf{v}_{y,i}$, and $\mathbf{\Sigma}$ a diagonal matrix containing the singular values σ_i in decending order. The first left and right singular vectors correspond to the maximum covariance, given by the largest singular value σ_1 .

Since the pattern vectors are unitary, datasets can be expanded as

$$\mathbf{Z}_s = \mathbf{P}_s \mathbf{V}_s^H, \quad (1.34)$$

with the expansion coefficients $\mathbf{P}_s$. The expansion coefficients $\mathbf{P}_x$ and $\mathbf{P}_y$ are uncorrelated with each other:

$$\mathbf{P}_x^H \mathbf{P}_y = \mathbf{V}_x^H \mathbf{Z}_x^H \mathbf{Z}_y \mathbf{V}_y \stackrel{(1.32)}{=} (N-1) \mathbf{V}_x^H \mathbf{C}_{xy} \mathbf{V}_y \stackrel{(1.33)}{=} (N-1) \mathbf{\Sigma}. \quad (1.35)$$

However, the expansion coefficients $\mathbf{P}_x$ and $\mathbf{P}_y$ are generally correlated within themselves as indicated by the condition $\mathbf{P}_s^H \mathbf{P}_s \neq c \mathbf{I}_M$ for any arbitrary constant c .

The total covariance, given by the squared Frobenius norm of the cross-covariance matrix, is

$$\|\mathbf{C}_{xy}\|_F^2 = \sum_{i=1}^M \sigma_i^2, \quad (1.36)$$

which shows that each mode captures a fraction of the total squared covariance. The relative importance of each mode can be quantified by the squared covariance fraction $f_{xy,i}$:

$$f_{xy,i} = \frac{\sigma_i^2}{\sum_{i=1}^M \sigma_i^2}. \quad (1.37)$$

This fraction provides a measure of how much of the shared variability between the two datasets is captured by the i^{th} mode. To assess the importance of the patterns themselves, it is necessary to consider the variance within the individual datasets explained by each mode relative to the total variance of the individual datasets. This can be expressed using the squared Frobenius norm of the individual datasets:

$$f_{xx,i} = \frac{\mathbf{v}_{x,i}^H \mathbf{C}_{xx} \mathbf{v}_{x,i}}{\|\mathbf{Z}_x\|_F^2} = \frac{1}{N-1} \frac{\mathbf{p}_{x,i}^H \mathbf{p}_{x,i}}{\|\mathbf{Z}_x\|_F^2}, \quad (1.38)$$

$$f_{yy,i} = \frac{\mathbf{v}_{y,i}^H \mathbf{C}_{yy} \mathbf{v}_{y,i}}{\|\mathbf{Z}_y\|_F^2} = \frac{1}{N-1} \frac{\mathbf{p}_{y,i}^H \mathbf{p}_{y,i}}{\|\mathbf{Z}_y\|_F^2}. \quad (1.39)$$

While the squared covariance fraction $f_{xy,i}$ indicates the amount of shared variability between the two datasets, it does not provide information about the strength of the association or coupling between them. To measure this, we use the correlation coefficient ρ_i for the expansion coefficients of mode i between the two datasets:

$$\rho_i = \frac{\text{cov}(\mathbf{p}_{1,i}, \mathbf{p}_{2,i})}{(\text{var}(\mathbf{p}_{1,i}) \text{var}(\mathbf{p}_{2,i}))^{1/2}} = \frac{\sigma_i}{\|\mathbf{p}_{1,i}\|^2 \|\mathbf{p}_{2,i}\|^2}. \quad (1.40)$$

The homogeneous patterns [7] for the pattern vectors are:

$$\text{HoP}_{x,i} = \text{cov}(\mathbf{Z}_x, \mathbf{p}_{x,i}) = \mathbf{S}_{xx} \mathbf{v}_{x,i}, \quad (1.41)$$

$$\text{HoP}_{y,i} = \text{cov}(\mathbf{Z}_y, \mathbf{p}_{y,i}) = \mathbf{S}_{yy} \mathbf{v}_{y,i}, \quad (1.42)$$

These patterns provide an indicator of which variables or features are important for the covarying components in both fields. The corresponding heterogeneous patterns are defined as:

$$\text{HeP}_{x,i} = \text{cov}(\mathbf{Z}_x, \mathbf{p}_{y,i}) = \sigma_i \mathbf{v}_{y,i}, \quad (1.43)$$

$$\text{HeP}_{y,i} = \text{cov}(\mathbf{Z}_y, \mathbf{p}_{x,i}) = \sigma_i \mathbf{v}_{x,i}. \quad (1.44)$$

These patterns indicate how well the variables in one field can be predicted from the i^{th} expansion coefficient of the other field. One may also use standardized data, $\mathbf{Z}_x$ and $\mathbf{Z}_y$, to obtain the respective homogeneous and heterogeneous correlation patterns.

MCA gained popularity in the field of meteorology¹¹ through the works of Bretherton *et al.* [7] and Wallace *et al.* [76], since when it has been extensively applied [e.g. 3, 52, 60, 65, 75, 76].

The method's orthogonal patterns ensure efficient dimensionality reduction of the covariance but limit physical interpretability, similar to PCA. To enhance interpretability, rotation of the MCA solution can be employed, as proposed by Cheng & Dunkerton [17], and later applied by Frankignoul & Kestenare [28]. A composite matrix of the left and right singular vectors is constructed and a simplicity criterion (e.g. varimax) is applied:

$$\mathbf{V}_R = \begin{bmatrix} \mathbf{V}_x \\ \mathbf{V}_y \end{bmatrix} \mathbf{M} \quad (1.45)$$

In this case, $\mathbf{M}$ is a diagonal matrix that weights the augmented pattern matrix. While Cheng & Dunkerton [17] proposed $\mathbf{M} = \Sigma^{1/2}$, this choice is not ideal as the singular values σ_i are not directly related to the explained covariance of the patterns. Instead, it is more appropriate to weight each mode by the squared singular value, preserving the squared covariance fraction of the first m modes before and after (orthogonal) rotation. Thus, we use

¹¹According to Jolliffe [44, p. 225], the technique was first used in psychology in the 1950s under the name *inter-battery factor analysis*. As a variant of Partial Least Squares (PLS), it was later applied in biomedical sciences by Sampson *et al.* [69] and was referred to as *PLS-SVD* by Wegelin [79]. In meteorology, Bretherton *et al.* [7] introduced the term *SVD analysis*, which remains in use today [e.g. 60]. Occasionally, the method is also called *Canonical Covariance Analysis*, similar to Canonical Correlation Analysis [e.g. 19, 35].

$$M = \Sigma \quad (1.46) \quad 1$$

for the weighted (or "loaded") pattern vectors. 2

Beyond rotation for better interpretability, Hilbert MCA can be applied to capture the temporal evolution of coupled patterns, akin to Hilbert PCA (see Section 1.1.3). This is especially useful for analyzing propagating patterns like the MJO, which involve covarying patterns in different climate variables, such as wind speed in the troposphere and lower stratosphere, and precipitation. By augmenting data with the Hilbert transform, we obtain complexified fields and solve the maximization problem (Equation (1.31)) for these augmented data, capturing the temporal evolution of coupled patterns. It seems that Hilbert MCA has seen limited application in climate science [e.g. 25]. In this thesis we utilize the combined power of Hilbert and rotated MCA to identify and interpret coupled patterns of variability in the climate system. 11

PCA, MCA and CCA MCA reduces to PCA when both sets are identical. Furthermore, MCA can be viewed as two PCAs followed by rotation to align the results of the two analyses as closely as possible [20]. MCA is also closely related to Canonical Correlation Analysis (CCA), which, instead of maximizing shared covariance, seeks patterns that maximize correlation. Unlike MCA, where pattern vectors are by construction orthogonal, yielding correlated expansion coefficients, CCA results in pattern vectors that are not necessarily orthogonal but produce uncorrelated canonical variates [44, 80]. In fact, PCA, MCA and CCA can all be seen as special cases of a more general framework [borga_unified_1997, 72]. 19

1.3 | Neural Networks 20

1.3.1 Introduction to Neural Networks 21

Definition and Overview: Explain what neural networks are, including a brief history and their importance in machine learning. Basic Architecture: Describe the basic structure of a neural network, including neurons, layers (input, hidden, output), activation functions, and the learning process. 25

1.3.2 Recurrent Neural Networks

Introduction to RNNs¹²: Describe the architecture of RNNs, their ability to handle sequential data, and their applications. Challenges with RNNs: Explain issues such as vanishing and exploding gradients.

Long Short-Term Memory (LSTM) Networks

Introduction to LSTM¹³: Discuss the motivation behind LSTM networks and how they address the limitations of standard RNNs. LSTM Architecture: Describe the components of LSTM cells (input gate, forget gate, output gate) and their functions. Mathematical Formulation: Provide the equations governing LSTM networks. Applications of LSTM: Highlight key applications where LSTMs have shown significant performance improvements.

1.3.3 Explainable AI Techniques

Introduction to Explainable AI: Explain the need for explainable AI, particularly in critical applications where understanding model decisions is crucial. Overview of Techniques: Briefly list various explainable AI methods, with a focus on LRP and SHAP.

Layerwise Relevance Propagation (LRP)

Concept of LRP¹⁴: Describe what LRP is and its importance in understanding neural network decisions. Mechanism of LRP: Explain how LRP works, including the propagation rules and the relevance score computation. Applications of LRP: Provide examples of how LRP is used in practice to interpret model predictions.

simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables. simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables

SHapley Additive exPlanations (SHAP)

Introduction to SHAP¹⁵: Discuss the SHAP method, its foundation in cooperative game theory, and its advantages. SHAP Values: Explain how SHAP values are computed and

¹²Graves 2021 Supervised Sequence Labelling with Recurrent Neural Networks

¹³Hochreiter & Schmidhuber 1997

¹⁴Bach et al. 2015

¹⁵Lundberg 2017

their interpretation. Applications of SHAP: Provide examples of SHAP in various domains, highlighting its ability to provide consistent and locally accurate explanations.

1.4 | Gaussian Processes

Class of pattern recognition techniques that typically require a metric to be defined that measures the similarity of any two vectors in input space, and are generally fast to "train" but slow at making predictions for test data points.

1.4.1 Introduction to Gaussian Processes

Definition and Overview: Explain what Gaussian Processes are, their significance in machine learning, and their use in modeling continuous data. Basic Concepts: Introduce key concepts such as Gaussian distributions, covariance functions, and kernels.

1.4.2 Gaussian Processes for Regression

GP Regression Model: Describe the formulation of GP for regression tasks, including prior, likelihood, and posterior distributions. Kernel Functions: Discuss common kernel functions (e.g., squared exponential, Matérn) and their role in defining the GP's properties. Hyperparameter Tuning: Explain how to tune hyperparameters using techniques like maximum likelihood estimation.

1.4.3 Spatial Modeling with Gaussian Processes

Spatial Data: Define spatial data and the importance of modeling spatial dependencies. GPs in Spatial Modeling: Explain how GPs are particularly suited for spatial data due to their ability to model complex, non-linear relationships and provide uncertainty estimates.

Log Gaussian Cox Process

Introduction to Cox Processes: Explain Cox processes and their relevance in spatial point process modeling. Log Gaussian Cox Model: Describe the log Gaussian Cox model, which assumes the log intensity function is modeled by a GP. Mathematical Formulation: Provide the mathematical formulation of the log Gaussian Cox model.

Negative Binomial Distribution

Introduction to Negative Binomial Distribution: Explain the negative binomial distribution and its use in modeling overdispersed count data. Integration with Log Gaussian Cox Model: Describe how the negative binomial distribution can be integrated into the log Gaussian Cox model to handle count data with overdispersion.

1.4.4 Model Implementation and Inference

Model Fitting: Discuss methods for fitting the log Gaussian Cox model with a negative binomial distribution, including Markov Chain Monte Carlo (MCMC) and variational inference. Inference and Prediction: Explain how to perform inference and make predictions using the fitted model.

1.5 | Comparative Perspective

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Proofs

The Frobenius norm $\|\cdot\|_F$ of a complex $N \times P$ matrix $\mathbf{A} \in \mathbb{C}^{N \times P}$ is defined as:

$$\|\mathbf{A}\|_F \equiv \left(\sum_i^N \sum_j^P |\alpha_{ij}|^2 \right)^{1/2} = \left(\sum_i^{\min(N,P)} \sigma_i^2(\mathbf{A}) \right)^{1/2}. \quad (\text{A.1})$$

This expression reveals that the Frobenius norm is equivalent to the ℓ_2 -norm of the matrix's singular values σ_i . We can demonstrate this equivalence using the singular value decomposition (SVD) of $\mathbf{A}$ and leveraging the property that the trace of a matrix remains invariant under cyclic permutations.

Starting with the definition of the Frobenius norm, we have:

$$\|\mathbf{A}\|_F = \left(\text{tr}(\mathbf{A}^H \mathbf{A}) \right)^{1/2}$$

Applying the SVD, we can express $\mathbf{A} = \mathbf{V} \mathbf{\Sigma} \mathbf{V}^H$, leading to:

$$\|\mathbf{A}\|_F = \left(\text{tr}(\mathbf{V} \mathbf{\Sigma} \mathbf{U}^H \mathbf{U} \mathbf{\Sigma}^H \mathbf{V}^H) \right)^{1/2}$$

Since $\mathbf{U}^H \mathbf{U} = \mathbf{I}$, this simplifies to:

$$\|\mathbf{A}\|_F = \left(\text{tr}(\mathbf{V} \mathbf{\Sigma}^2 \mathbf{V}^H) \right)^{1/2}$$

Finally using the invariance property of the trace, we obtain:

1

$$\|\mathbf{A}\|_{\text{F}} = \left(\sum_i^{\min(N,P)} \sigma_i^2(\mathbf{A}) \right)^{1/2}$$

- 2 This concludes the proof that the Frobenius norm of a matrix is indeed the ℓ_2 -norm of its
3 singular values.

